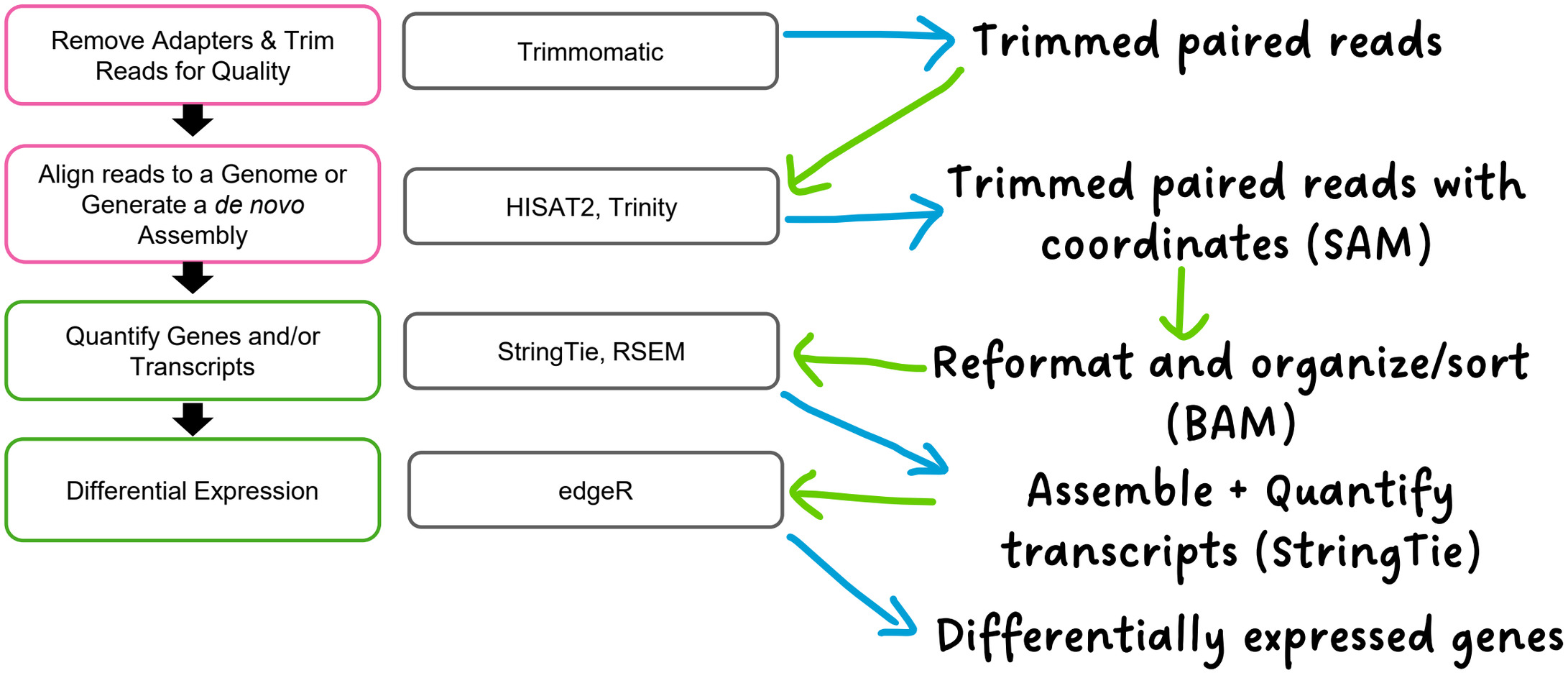
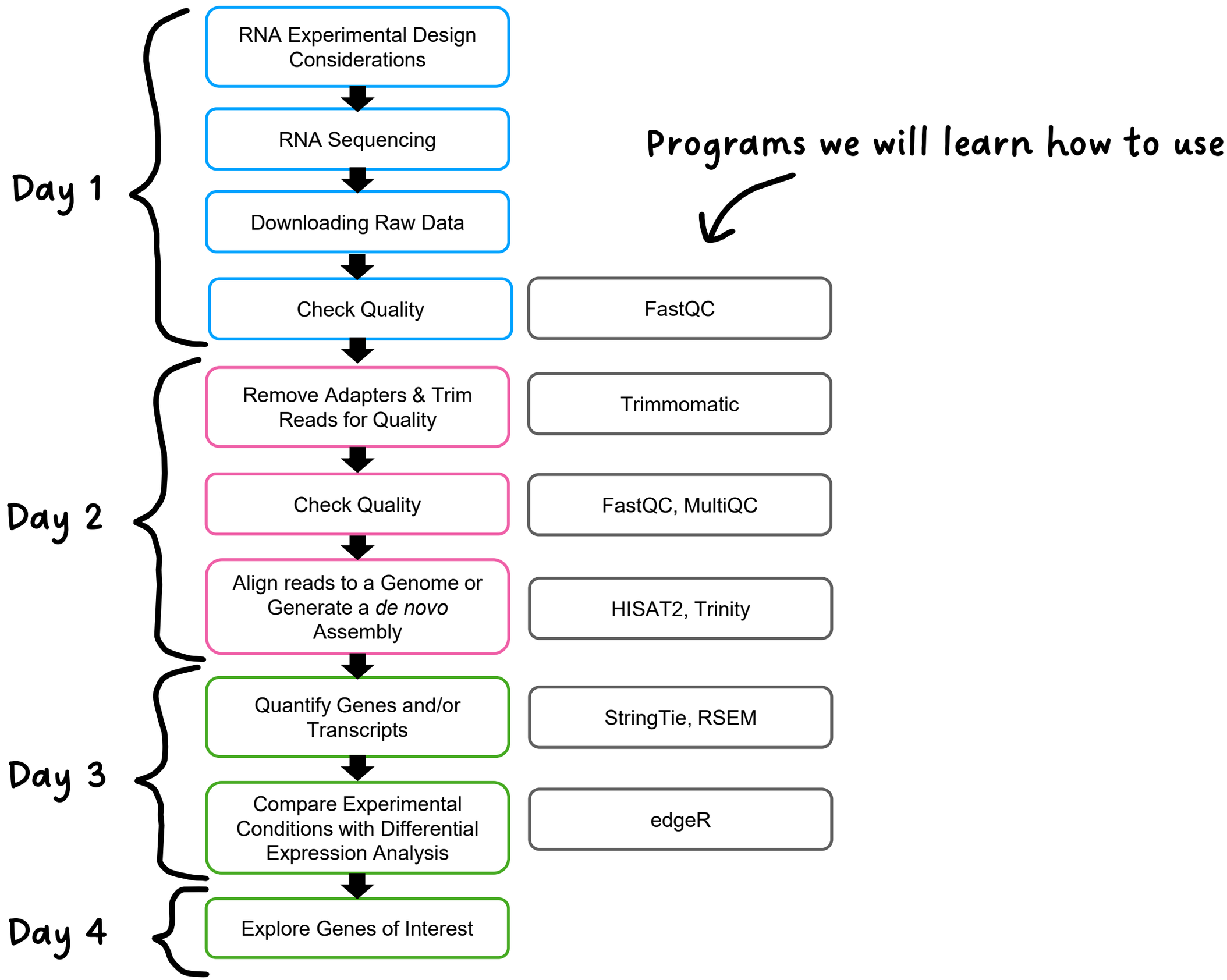
RNA-seq and analysis of differential gene expression

Slides adapted from: Tom Booker & Bernadette Johnson
Bioinformatics for Evolutionary Biology and Conservation

Pipeline Recap



Basic RNA-seq pipeline



Let's check our files from last time

Did everything run according to plan?

1. Did we generate count matrix from StringTie?

Should see the gene and the count for each sample, separate by a comma.

```
ubuntu@bioinformatics-1-do-not-delete:~/05_stringtie$ head gene_count_matrix.csv
gene_id,C4NPP,C7NPP,C7PP,C88NPP,C8PP,C9NPP,C9PP
MSTRG.3|ndufc1,3,4,0,0,0,0,0
MSTRG.1|LOC125980798,8,7,0,0,0,10,0
MSTRG.2|LOC125987402,0,0,0,0,0,0,0
MSTRG.4|hars,7,0,0,0,4,9,0
MSTRG.5|LOC137840633,0,0,0,0,0,0,0
gene-slc7a11|slc7a11,4,0,0,0,0,0,0
gene-LOC125981049|LOC125981049,0,0,0,0,0,0,0
gene-LOC137840640|LOC137840640,0,0,0,0,0,0,0
gene-LOC137840637|LOC137840637,0,0,0,0,0,0,0
```

Normalization

- Why is it important to normalize RNA-seq counts?
- How does edgeR work?

Why is it important to normalize?

Differences between samples

Differences between genes

Sequencing depth

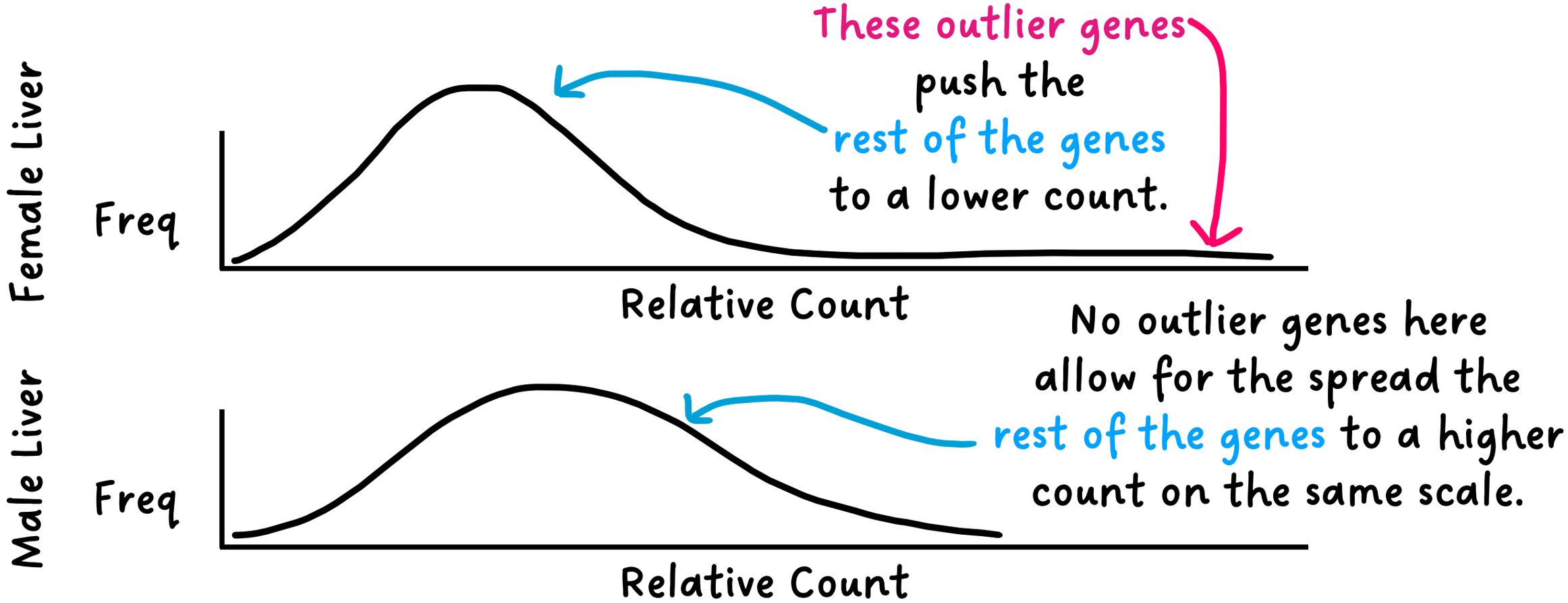
Garber et al. 2011

Why is it important to normalize?

Differences between samples

Differences between genes

Compositional bias

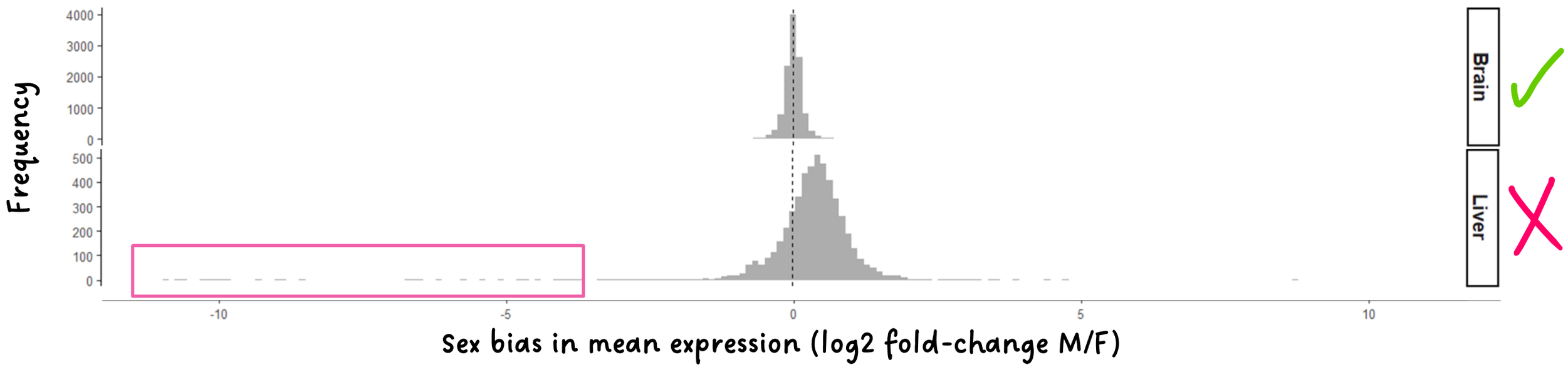


Remember RNA-seq is a relative measurement!

Why is it important to normalize?

Compositional bias

Real world example:



How does edgeR work?

1. **Filter low-count genes**: remove noisy low count genes, this improves the stability of variance estimates
2. **Normalize library sizes (TMM)**: correct compositional bias + seq. depth
3. **Construct design matrix**: which samples are in which test group
4. **Estimate dispersions**: to quantify biological variability across replicates, (estimates common, trended, and tagwise dispersion)
5. **Fit quasi-likelihood GLM**: models gene expression while accounting for dispersion uncertainty
6. **Hypothesis testing**: test whether expression differs between test groups
7. **Multiple testing correction**: Benjamini–Hochberg FDR to control for false discovery rate

Let’s take a look at our data

We will be working with a dataset generated by running:
Trimmomatic > HISAT2 > StringTie

In the workshop directory you will have a counts matrix generated from a full dataset, that looks like this:

```
ubuntu@bioinformatics-1-do-not-delete:~/genecounts$ head gene_count_matrix.csv
gene_id,Control1_NP,Control2_NP,Control3_NP,Control4_NP,Control1_P,Control2_P,Control3_P,Control4_P
MSTRG.7100,14293,11145,11462,9558,10823,9095,9817,24719,28988,17402,11246,14971
MSTRG.9818,513,348,412,278,335,393,315,1037,948,354,406,425
MSTRG.15559,277,187,166,289,158,136,167,593,410,186,168,179
MSTRG.8960,2517,2578,2343,1885,1839,1636,1314,5064,4463,2381,1798,2889
MSTRG.20088,6,3,6,4,5,0,0,0,10,8,14,0
MSTRG.20089,0,13,0,12,10,0,13,15,19,8,30,0
MSTRG.20086,2863,3524,2722,2027,1935,1861,1906,5786,5788,3260,2516,2950
MSTRG.20087,0,0,0,0,0,0,0,0,17,6,16,0
MSTRG.20084,0,0,0,0,4,21,6,12,0,0,0,0
```

But how do we take these counts and have biologically meaningful results?

First, we need to know what the data is representing.

Let's take a look at our data

The data we will be working with is from Gulf pipefish.

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Pipefish are relatives of seahorses, and seadragons.

Similarly to seahorses, males carry the offspring in a complex structure called a brood pouch.

The male provisions a lot of resources during offspring development.

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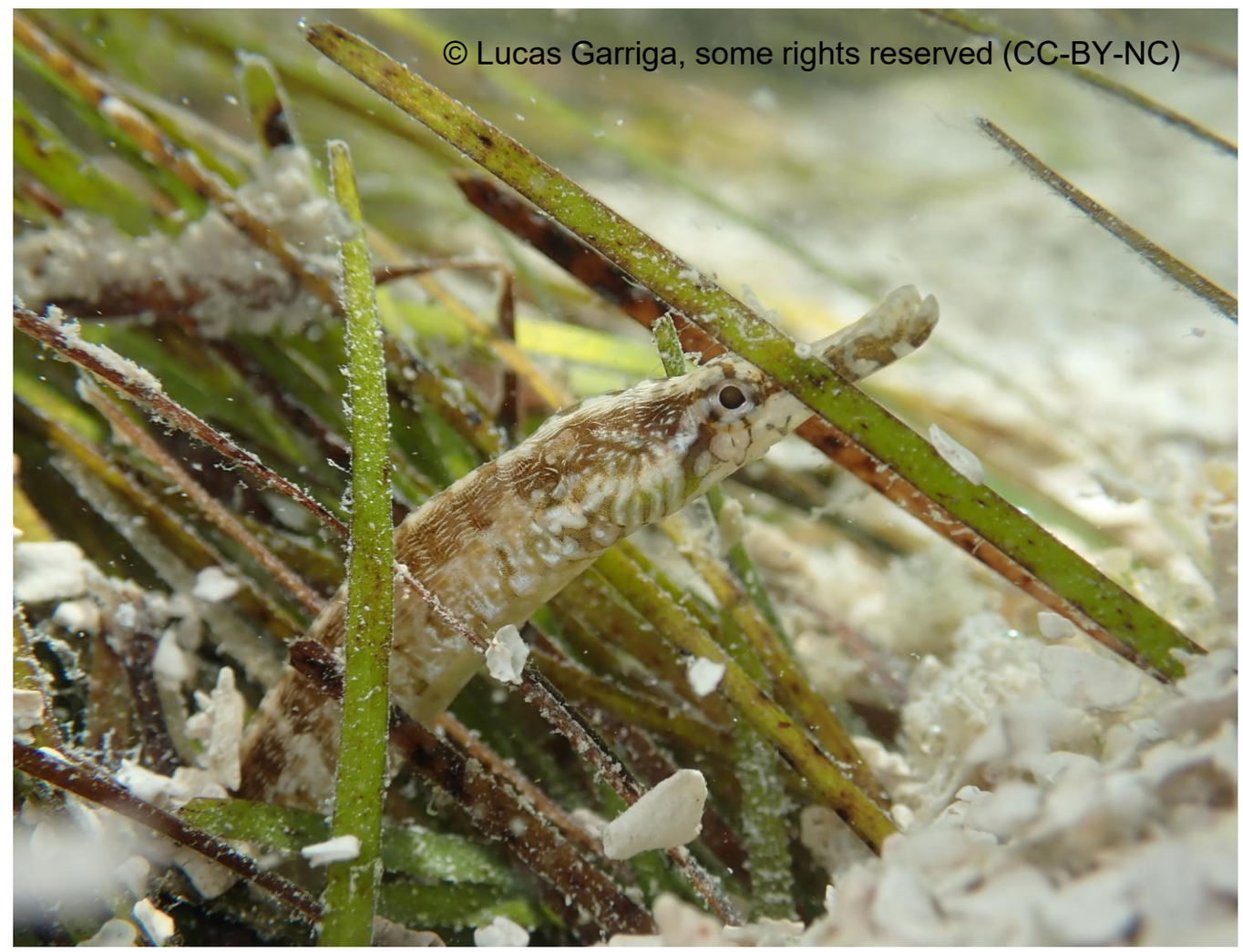
Let’s take a look at our data

Pipefish predominately live in undisturbed seagrass beds.



These habitats are particularly sensitive to pollution, especially from waste treatment facilities.

One major concern is that environmental estrogens (from human contraceptives) will impair their ability to reproduce.

Let's take a look at our data

To study this concern, scientists have collected samples, and generated RNA-seq data of:

1. Control Pregnant Males
2. Control Non-Pregnant Males
3. Estrogen-exposed Pregnant Males

This gives us the ability to test multiple comparisons with a differential expression test.

Pregnant vs Not Pregnant
Control Pregnant vs EE Pregnant

Let's take a look at our data

Now we are going to work in R studio!

The script is on the workshop GitHub.

1.4 Quick start

```
> x <- read.delim("TableOfCounts.txt",row.names="Symbol")
> group <- factor(c(1,1,2,2))
> y <- DGEList(counts=x,group=group)
> keep <- filterByExpr(y)
> y <- y[keep,,keep.lib.sizes=FALSE]
> y <- normLibSizes(y)
> design <- model.matrix(~group)
> fit <- glmQLFit(y,design)
> qlf <- glmQLFTest(fit,coef=2)
> topTags(qlf)
```

bioconductor.org/packages/release/bioc/vignettes/edgeR/inst/doc/edgeRUsersGuide.pdf

What to do after we have a DEGs?

If I don’t have gene annotations, how do I know what these genes are?

```
> summary(decideTests(qlf))
       group_pairCNP
Down             104
NotSig         17669
Up               122
```

What to do after we have a DEGs?

	logFC	logCPM	F	PValue	FDR
MSTRG.19531	7.160553	7.832505	1311.925	2.52E-11	4.50E-07
MSTRG.19535	7.28065	7.873645	824.5068	2.15E-10	1.92E-06
MSTRG.20576	7.433213	10.17989	516.3945	1.81E-09	1.08E-05
MSTRG.18768	-2.48302	6.650381	313.7864	1.75E-08	7.83E-05
MSTRG.20560	-3.19692	3.48937	224.2389	8.09E-08	0.000289
MSTRG.18326	-2.60679	8.058441	159.972	3.58E-07	0.001068
MSTRG.14495	-1.42012	5.190076	119.5841	1.28E-06	0.003279
MSTRG.2148	-2.88172	5.363254	111.3908	1.75E-06	0.003397
MSTRG.747	-3.67026	1.881379	108.6425	1.99E-06	0.003397

We can take the StringTie IDs from our DEGs
Ex. MSTRG.19531

We probably want the sequence for that gene

Then we can find out what the gene is by doing a BLAST search on NCBI.

What to do after we have a DEGs?

MSTRG.19531
MSTRG.19535
MSTRG.20576
MSTRG.18768
MSTRG.20560
MSTRG.18326
MSTRG.14495
MSTRG.2148
MSTRG.747

You can take the merged_transcripts.gff output from StringTie along with gffread, you can extract transcript sequences (or you can directly use the *de novo* assembly from Trinity).

https://ccb.jhu.edu/software/stringtie/gff.shtml#gffread

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What to do after we have a DEGs?

MSTRG.19531
MSTRG.19535
MSTRG.20576
MSTRG.18768
MSTRG.20560
MSTRG.18326
MSTRG.14495
MSTRG.2148
MSTRG.747

```
>MSTRG.19531
ATGCGATCGACTACGCGAGCATCA
GCAGCTACGGACTAGCACGT...
>MSTRG.19535
ATGCGATCGACTACGCGAGCATCA
GCAGCTACGGACTAGCACGT...
```

We can take the StringTie IDs from our DEGs Ex. MSTRG.19531

We probably want the sequence for that gene

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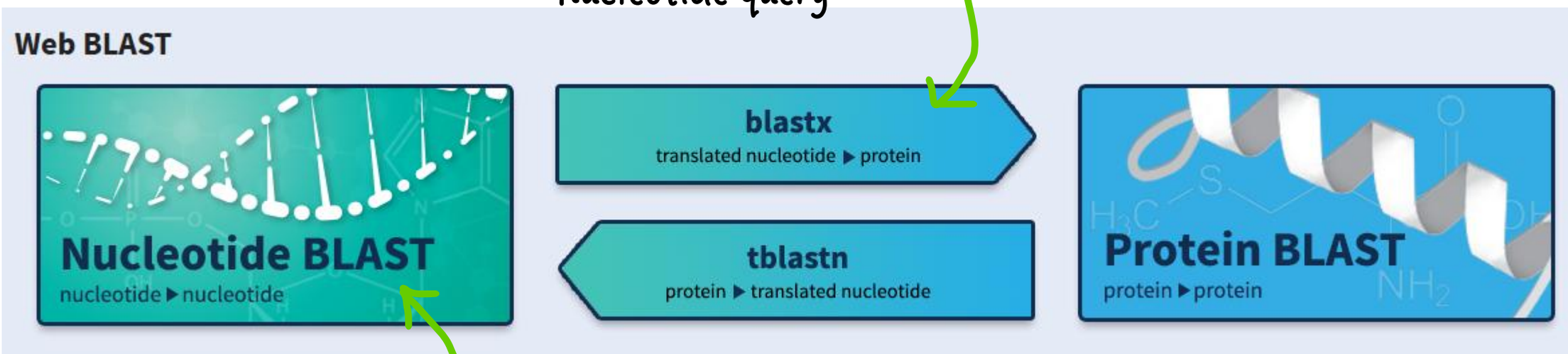
What to do after we have a DEGs?

Search for protein sequences using translated nucleotide query



https://blast.ncbi.nlm.nih.gov/Blast.cgi

Search for nucleotide sequences using nucleotide query

We can take the StringTie IDs from our DEGs Ex. MSTRG.19531

We probably want the sequence for that gene

Then we can find out what the gene is by doing a BLAST search on NCBI.

What to do after we have a DEGs?

What if you want to do a GO-type analysis (with a species with no annotation)? First determine if you can find a species with a shallow enough divergence time from your species (this can be a little subjective) that has GO information available.

Then take your sequences (nucleotides), and we will probably want to compare at a protein level (amino acid) instead of a nucleotide level.

Convert the nucleotide sequence to its longest open reading frame.

(fastatoorf)

https://github.com/JonesLabIdaho/FastaToORF

Search for these sequences in a second species annotation file (reciprocal search) to find Orthologs.

https://www.ncbi.nlm.nih.gov/guide/howto/run-blast-local/
https://github.com/eggnogdb/eggnog-mapper

What to do after we have a DEGs?

Beginner Friendly Tools to Understand Your DEGs

PANTHER Classification System: online database where you can enter IDs from one of their supported species, and perform a series of analyses (functional classification, overrepresentation, or enrichment test) (https://pantherdb.org/)

Kyoto Encyclopedia of Genes and Genomes (KEGG): focuses more on pathway analysis (https://www.genome.jp/kegg/)

Gene ontology (GO): provides different levels of annotation and ontology (https://geneontology.org/)

ShinyGO: provides annotation + nice R visuals (https://bioinformatics.sdstate.edu/go/)

String: provides protein-protein interaction network analyses (https://string-db.org/)

Course Info

- Course Materials on Github: https://github.com/BernadetteBiology
- How to install and run popular RNA-seq analysis programs: https://bernadettebiology.weebly.com/protocols-tutorials--courses.html
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